

Supervisor Summary Report

AI-Driven M&A Synergy Prediction via Heterogeneous Graph Neural Networks

Student: Hard Joshi
Degree: Data Science & AI
Supervisor: Arish Siddiqui

ID: 2512658
Module: CN6000
Date: 8 May 2026

Problem Statement

M&A transactions destroy shareholder value in 70–90% of cases. Despite this, existing predictive models consistently fail to identify value-destroying deals before announcement. This study argues the failure is architectural: every prior quantitative M&A model treats firms as isolated balance-sheet vectors, blind to the supply-chain networks, competitive topologies, and section-level semantic signals that actually determine whether a deal will create or destroy value.

Research Design

A tri-modal late-fusion architecture (**HeteroGraphSAGE**) was constructed and evaluated on 3,000 completed US domestic M&A transactions (2000–2023), sourced from Yahoo Finance, SEC EDGAR, and Bloomberg SPLC.

Block	Modality	Description
A	Financial	56 ratio features: leverage, liquidity, profitability, deal structure
B	Textual	FinBERT embeddings of MD&A and Risk Factors from pre-deal 10-K filings, kept section-separate to preserve opposing economic signals
C	Graph	Two-hop heterogeneous GraphSAGE embeddings from Bloomberg SPLC supply-chain network

Three formal hypotheses were tested against Bonferroni-corrected thresholds ($\alpha = 0.0167$):

- **H1 — Topological Alpha:** Graph topology adds predictive signal beyond financial features.
- **H2 — Semantic Divergence:** MD&A and Risk Factor sections encode economically opposing signals.
- **H3 — Topological Arbitrage:** Network centrality compresses announcement-return variance.

Evaluation used five-fold stratified cross-validation with strict temporal splits (train: 2000–2016 / val: 2017–2019 / test: 2020–2023) and an 11-day event-window embargo to prevent leakage.

Key Empirical Results

Classification Pipeline (AUC-ROC — Direction Prediction)

Model	Configuration	AUC	Acc.	F1
M1	Financial only	0.5408	52.8%	0.473
M2	Financial + Text	0.5289	52.9%	0.476
M3	Full Fusion	0.5655	54.8%	0.490
M3e	M3 + Aux Feats.	0.5585	55.1%	0.492

The headline finding: **M3 achieves AUC = 0.5655, a +0.0247 lift over the financial-only baseline.** Hyperparameter tuning alone did not improve any model, confirming that architectural signal choice matters more than optimiser search.

A key negative result: M2 (Financial + naive text) **reduces** AUC by -0.0119 versus M1. This is the M2 Reversal — adding undifferentiated FinBERT text actively destroys predictive value when MD&A and Risk Factor signals are pooled without section-separation.

Regression Pipeline (Continuous CAR Magnitude)

All regression configurations produced $R^2 < 0$, meaning the model cannot explain continuous CAR magnitude better than the sample mean. This is an honest and informative negative result: short-window announcement returns are dominated by idiosyncratic shocks (competing bids, investor sentiment, payment structure) that are structurally invisible in pre-announcement data. The boundary between tractable (direction) and intractable (magnitude) prediction sub-problems is precisely located.

Hypothesis Test Verdicts

H	Claim	Evidence	Verdict
H1	Topology adds signal	AUC lift +0.0247; paired t -test, $p < 0.0167$	✓ Supported
H2	Section semantics diverge	$\beta_{\text{MDA}} = +0.0044$, $\beta_{\text{RF}} = -0.0080$; OLS $p < 0.0167$	✓ Supported
H3	Centrality dampens variance	$F_{\text{Levene}} = 7.07$, $p = 0.0079$; $r = -0.0701$, $p = 0.0002$	✓ Supported

Research Contributions

1. **Methodological:** One of the first explicit applications of heterogeneous GNNs to post-merger CAR classification, formally moving the field beyond the tabular independence assumption.

2. **Empirical:** Establishes that section-level semantic divergence between MD&A and Risk Factor disclosures is directional and economically meaningful in M&A outcome prediction. Conflating these sections is demonstrably harmful.
3. **Practical Artefact:** The **Deal Intelligence Terminal** — a fully interactive research dashboard rendering all model outputs, ablation ladders, SHAP decompositions, and hypothesis tests dynamically — ensures complete reproducibility.

Practical Significance of +0.025 AUC

A +0.025 AUC improvement means the model is 2.5 percentage points more likely than chance to correctly rank a value-creating deal above a value-destroying one when comparing random deal pairs. In advisory practice, M&A evaluation is a ranking problem: teams need a better triage of which opportunities deserve deeper diligence, not perfect foresight. Applied across a large deal pipeline, this persistent edge translates into systematically better capital allocation.

Key Limitations

- The headline AUC of 0.5655 is meaningful for research but not commercially deployable as a stand-alone decision system.
- The text pipeline uses PCA compression of section embeddings; a cross-attention dual-stream encoder would likely recover additional semantic signal.
- H3 is a structural finding: high-centrality firms may also be larger and more analyst-covered, so centrality cannot be claimed as an exhaustively isolated causal driver.
- The sample is US-listed firms only; generalisability to cross-border or private-equity transactions remains untested.

Handbook Compliance Summary (CN6000)

Criterion	Met
Significant non-trivial computing problem solved	✓
Literature review analysing prior work	✓
Specification, design, and implementation of a system	✓
Ethical, legal, and social considerations addressed	✓
Evaluation of own project work against research of others	✓
Harvard (Cite Them Right) referencing	✓
Ethical approval submitted	✓
Personal reflection and Mental Wealth section	✓
Interactive artefact (Deal Intelligence Terminal)	✓
10,000 word report	✓

This summary was prepared alongside the full dissertation submitted for CN6000 on 8 May 2026.
*Full report: **cn6000_final_submission.pdf***